

EXECUTIVE SUMMARY

Target: http://dedup-test.com
Scan ID: TORTURE-P11-cc8168ae
Scan Date: 2026-03-05 16:14:56
Scan Duration: N/A
Total Requests Sent: 33
Avg Latency: N/A
Peak Concurrency: N/A
Findings: 3 vulnerabilities detected

Severity Breakdown: Critical: 0 | High: 1 | Medium: 2 | Low: 0
AI Inference Calls: 0
LLM Avg Latency: N/A
Circuit Breaker Activations: 0

- Executive Summary:
- Overall Risk Level: High due to a single high-severity vulnerability identified in system infrastructure.
- Most Critical Category: HIGH (1) - Requires urgent remediation to prevent potential data breaches or service disruptions.
- Immediate Actions: Prioritize patching the identified high-risk vulnerability and implement additional security controls for mitigation.

EXECUTIVE STRATEGIC IMPACT

[SYSTEM]: CortexEngine
Strategic Attack Path Analysis: An adversary could exploit BROKEN_AUTH by leveraging authentication bypass techniques, then use the resulting unauthorized access to inject malicious XSS payloads into vulnerable web inputs. By chaining these vulnerabilities together, an attacker would gain initial foothold through broken auth mechanisms and subsequently execute cross-site scripting attacks to escalate privileges or exfiltrate sensitive data across

DETAILED FINDINGS

FILTER: AUTHENTICATION GATES

Finding #1: Broken Authentication

CRITICAL

CWE: CWE-287
CVSS Score: 10.0 (CRITICAL)

THREAT SCORE: 100/100

Description:

- Vulnerability 'Broken Authentication' detected in target endpoint.

Impact:

- Unauthorized access to system resources confirmed.

Explanation:

Agent Gamma flagged this interaction based on high-confidence heuristic anomalies. Pattern 'BROKEN_AUTH' was intercepted to protect system integrity.

FORENSIC ANALYSIS

Method: GET | Param: param_0
URL: http://test-target-torture-.com/api/v1/endpoint_0
Analysis: The application failed to validate and sanitize user input, allowing arbitrary payload injection.

Table 1: Payload Decomposition

Component	Value	Technical Function
param_0	test_payload_0	Direct reference to target resource.
Access Check	Missing	Application fails to verify authorization.

Result	200 OK	Server returns data for unauthorized request.
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Payload Specifications:

Vector Category: Broken Authentication
Raw Payload: test_payload_0
Encoded: 746573745f7061796c6661645f30
Encoding Type: None

Reproduction Command:

```
curl -X GET 'http://test-target-torture-.com/api/v1/endpoint_0'
```

HTTP Traffic Snapshot:

Request:

GET http://test-target-torture-.com/api/v1/endpoint_0 HTTP/1.1
Host: target
{}

Response:

HTTP/1.1 200 OK
Content-Type: application/json

{"status": "vulnerable", "data": "revealed"}

Remediation:

- Implement strict input validation.

Recommended Code Fix:

```
from flask import Flask, request, jsonify
import jwt
import datetime

app = Flask(__name__)

SECRET_KEY = "your-secret-key"
ALGORITHM = "HS256"

def generate_token(user_id):
    payload = {
        'user_id': user_id,
        'exp': datetime.datetime.utcnow() + datetime.timedelta(hours=1)
    }
    token = jwt.encode(payload, SECRET_KEY, algorithm=ALGORITHM)
    return token

@app.route('/login', methods=['POST'])
def login():
```

```
data = request.get_json()
username = data.get('username')
password = data.get('password')

# Validate credentials (implementation omitted for brevity)

if not username or not password:
    return jsonify({'error': 'Invalid credentials'}), 401

user_id = get_user_id(username) # Placeholder function
token = generate_token(user_id)

return jsonify({'token': token}), 200
```

FILTER: INJECTION & FUZZING

Finding #2: Cross-Site Scripting (XSS)

HIGH

CWE: CWE-79
CVSS Score: 7.1 (HIGH)

THREAT SCORE: 71/100

Description:

- Vulnerability 'Cross-Site Scripting (XSS)' detected in target endpoint.

Impact:

- Unauthorized access to system resources confirmed.

Explanation:

Agent Gamma flagged this interaction based on high-confidence heuristic anomalies. Pattern 'XSS' was intercepted to protect system integrity.

FORENSIC ANALYSIS

Method: PUT | Param: param_1
URL: http://test-target-torture-.com/api/v1/endpoint_1
Analysis: The application failed to properly sanitize user input before rendering it in a web page.

Table 1: Payload Decomposition

Component	Value	Technical Function
param_1	test_payload_1	Direct reference to target resource.
Access Check	Missing	Application fails to verify authorization.
Result	200 OK	Server returns data for unauthorized request.

Payload Specifications:

Vector Category: Cross-Site Scripting (XSS)

Raw Payload: test_payload_1

Encoded: 746573745f7061796c6f61645f31

Encoding Type: None

Reproduction Command:

```
curl -X PUT 'http://test-target-torture-.com/api/v1/endpoint_1'
```

HTTP Traffic Snapshot:

Request:

PUT http://test-target-torture-.com/api/v1/endpoint_1 HTTP/1.1

Host: target

{}

Response:

HTTP/1.1 200 OK

Content-Type: application/json

{"status": "vulnerable", "data": "revealed"}

Remediation:

- Implement strict input validation.

Recommended Code Fix:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Secure Web Page</title>
  <script nonce="{{ csrf_token() }}">
    function loadScript(src) {
      var script = document.createElement('script');
      script.src = src;
      script.async = true;
      document.head.appendChild(script);
    }

    // Example usage
    loadScript('/path/to/secure-script.js');
  </script>
</head>
<body>

</body>
</html>
```

Finding #3: Cross-Site Scripting (XSS)

HIGH

CWE: CWE-79
CVSS Score: 7.1 (HIGH)

THREAT SCORE: 71/100

Description:

- Vulnerability 'Cross-Site Scripting (XSS)' detected in target endpoint.

Impact:

- Unauthorized access to system resources confirmed.

Explanation:

Agent Gamma flagged this interaction based on high-confidence heuristic anomalies. Pattern 'XSS' was intercepted to protect system integrity.

FORENSIC ANALYSIS

Method: POST | Param: param_2
URL: http://test-target-torture-.com/api/v1/endpoint_2
Analysis: The application failed to properly sanitize user input before rendering it in a web page.

Table 1: Payload Decomposition

Component	Value	Technical Function
param_2	test_payload_2	Direct reference to target resource.
Access Check	Missing	Application fails to verify authorization.
Result	200 OK	Server returns data for unauthorized request.

Payload Specifications:

Vector Category: Cross-Site Scripting (XSS)
Raw Payload: test_payload_2
Encoded: 746573745f7061796c6f61645f32
Encoding Type: None

Reproduction Command:

```
curl -X POST 'http://test-target-torture-.com/api/v1/endpoint_2'
```

HTTP Traffic Snapshot:

Request:

POST http://test-target-torture-.com/api/v1/endpoint_2 HTTP/1.1
Host: target
{}

Response:

HTTP/1.1 200 OK
Content-Type: application/json

{"status": "vulnerable", "data": "revealed"}

Remediation:

- Implement strict input validation.

Recommended Code Fix:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Secure Web Page</title>
  <script nonce="{{ csrf_token() }}">
    function loadScript(src) {
      var script = document.createElement('script');
      script.src = src;
      script.async = true;
      document.head.appendChild(script);
    }

    // Example usage
    loadScript('/path/to/secure-script.js');
  </script>
</head>
<body>

</body>
</html>
```


SCAN TIMELINE

[Orchestrator] TARGET_ACQUIRED - 2026-03-05 16:14:56

[Sigma] JOB_ASSIGNED - 2026-03-05 16:14:56

[Alpha] LOG - 2026-03-05 16:14:56

[Alpha] LOG - 2026-03-05 16:14:56

[Gamma] LOG - 2026-03-05 16:14:57

[Beta] LOG - 2026-03-05 16:14:57

[Beta] LOG - 2026-03-05 16:14:57

[Alpha] LOG - 2026-03-05 16:14:57

[Gamma] LOG - 2026-03-05 16:14:57

[Alpha] LOG - 2026-03-05 16:14:57

[Beta] LOG - 2026-03-05 16:14:57

[Beta] LOG - 2026-03-05 16:14:57

[Gamma] LOG - 2026-03-05 16:14:57

[Kappa] LOG - 2026-03-05 16:14:57

[Kappa] LOG - 2026-03-05 16:14:58

[Beta] LOG - 2026-03-05 16:14:58

[Alpha] LOG - 2026-03-05 16:14:58

[Kappa] LOG - 2026-03-05 16:14:58

[Beta] LOG - 2026-03-05 16:14:58

[Gamma] LOG - 2026-03-05 16:14:58

[Beta] LOG - 2026-03-05 16:14:58

[Alpha] LOG - 2026-03-05 16:14:58

[Gamma] VULN_CONFIRMED - 2026-03-05 16:14:58

[Gamma] VULN_CONFIRMED - 2026-03-05 16:14:58

[Gamma] VULN_CONFIRMED - 2026-03-05 16:14:59

[Gamma] VULN_CONFIRMED - 2026-03-05 16:14:59

[Kappa] GI5_LOG - 2026-03-05 16:15:06

[Gamma] VULN_CONFIRMED - 2026-03-05 16:14:59

[Gamma] VULN_CONFIRMED - 2026-03-05 16:14:59

[Gamma] VULN_CONFIRMED - 2026-03-05 16:14:59

[Gamma] VULN_CONFIRMED - 2026-03-05 16:14:59

[Kappa] GI5_LOG - 2026-03-05 16:15:06

[Kappa] GI5_LOG - 2026-03-05 16:15:06